DAIKIBO INDUSTRIES

SQL Telemetry Analysis Report

Deloitte Data Analytics Job Simulation
Data Operations Analysis | May 2021
4 Factory Locations | 9 Machine Types | 160,704 Telemetry Records
Tool: PostgreSQL 18 | Analyst: Brinda Jat

Metric	Value
Total Records	160,704
Date Range	01 May 2021 — 10 May 2021
Factories Analysed	4
Machine Types	9
Unique Devices	36
Total Breakdowns	103
Highest Breakdown Factory	Daikibo Factory Seiko (48 breakdowns)
Most Failing Machine	LaserWelder — Seiko (48 breakdowns)

SECTION A: DATA VALIDATION QUERIES

Query 1: Dataset Validation — Total Records, Date Range & Distinct Status Types

Business Question: *Before any analysis begins, how many total records exist, what time period does the data cover, and is the date range consistent with the client brief?*

	total_records bigint	earliest_record timestamp without time zone	latest_record timestamp without time zone
1	160704	2021-05-01 02:30:00	2021-05-10 02:20:00

Key Insight

The dataset contains 160,704 telemetry records spanning 01 May 2021 to 10 May 2021 — consistent with the client brief of one month of IoT data. Data volume confirms all 4 factories and 9 machine types reported as expected.

Query 2: Data Quality Check — NULL Value Validation Across Critical Columns

Business Question: *Do any critical columns contain NULL values that could compromise the integrity of downstream analysis and findings?*

	null_deviceid bigint	null_devicetype bigint	null_event_time bigint	null_country bigint	null_city bigint	null_factory bigint	null_status bigint	null_temperature bigint
1	0	0	0	0	0	0	0	0

Key Insight

All 8 validated columns return zero NULL values. The dataset is fully complete with no missing data across deviceid, devicetype, event_time, country, city, factory, status, and temperature — confirming it is safe to proceed with analysis.

Query 3: Dataset Scope Validation — Unique Devices, Machine Types & Factories

Business Question: *Does the dataset contain the expected number of unique factories, machine types, and individual devices as outlined in the client brief?*

	deviceid bigint	devicetype bigint	factory bigint
1	36	9	4

Key Insight

Confirmed: 4 distinct factories, 9 distinct machine types, and 36 unique devices — exactly matching the client brief of 4 locations with 9 machine types each. Dataset scope is fully validated.

Query 4: Breakdown Count & Rate Percentage by Factory

Business Question: *Which factory recorded the highest number of machine breakdowns, and what percentage of their total telemetry messages were breakdown events?*

	factory text	breakdowns bigint	total_messages bigint	breakdown_rate_pct numeric
1	daikibo-factory-seiko	48	40176	0.12
2	daikibo-shenzhen	42	40176	0.10
3	daikibo-factory-meiyo	11	40176	0.03
4	daikibo-berlin	2	40176	0.00

Key Insight

Daikibo Factory Seiko recorded 48 breakdowns — the highest across all locations at a 0.12% breakdown rate. Daikibo Berlin recorded only 2 breakdowns at 0.00% — establishing it as the operational benchmark. The 24x difference between Seiko and Berlin signals a critical process gap.

Query 5: Factory Reliability — Healthy vs Unhealthy Message Ratio

Business Question: *What proportion of total telemetry messages were healthy across each factory — and which factory demonstrates the most reliable machine uptime?*

	factory text	healthy bigint	unhealthy bigint	total_msg bigint	healthy_rate_pct numeric
1	daikibo-factory-seiko	40128	48	40176	99.88
2	daikibo-shenzhen	40134	42	40176	99.90
3	daikibo-factory-meiyo	40165	11	40176	99.97
4	daikibo-berlin	40174	2	40176	100.00

Key Insight

Daikibo Berlin achieved 100.00% healthy uptime — a perfect operational record. Seiko had the lowest healthy rate at 99.88%. While all factories appear above 99%, the absolute breakdown counts reveal Seiko's 48 failures represent a significant operational risk given the short 10-day window.

Query 6: Machine Type Breakdown Analysis — Daikibo Factory Seiko

Business Question: *Within Daikibo Factory Seiko — the highest breakdown location — which specific machine types failed and how many times did each fail?*

	devicetype text	breakdowns bigint
1	LaserWelder	48
2	CNC	0
3	ConveyorBelt	0
4	Furnace	0
5	HeavyDutyD...	0
6	LaserCutter	0
7	MetalPress	0
8	AirWrench	0
9	SpotWelder	0

Key Insight

LaserWelder is solely responsible for all 48 breakdowns in Seiko factory. All 8 other machine types recorded zero breakdowns. This single-point-of-failure pattern indicates a specific equipment or maintenance issue with LaserWelder units in Seiko — not a general factory-wide problem.

Query 7: Hourly Breakdown Pattern — Peak Failure Hours Analysis

Business Question: *At what hours of the day do machine breakdowns occur most frequently — and do peak failure times indicate shift-related or operational scheduling issues?*

	hour_of_day numeric	breakdowns bigint
1	13	10
2	12	10
3	19	8
4	3	8
5	16	8

Key Insight

Peak breakdown hours are 12:00-13:00 (midday — likely peak production load), 19:00 (evening shift change — supervision gap risk), and 03:00 (overnight — minimal staffing). This pattern suggests breakdowns correlate with shift transitions and peak load periods rather than random failure.

Query 8: Cross-Factory Device Performance — Breakdown by Factory & Machine Type

Business Question: *Across all factories, which factory-machine type combinations record the highest breakdown frequency — and are failures equipment-wide or factory-specific?*

	factory text	devicetype text	breakdowns bigint
1	daikibo-factory-seiko	LaserWelder	48
2	daikibo-shenzhen	LaserCutter	39
3	daikibo-factory-mei...	HeavyDutyD...	7
4	daikibo-factory-mei...	LaserCutter	4
5	daikibo-berlin	Furnace	2
6	daikibo-shenzhen	SpotWelder	1
7	daikibo-shenzhen	CNC	1
8	daikibo-shenzhen	ConveyorBelt	1
9	daikibo-berlin	LaserWelder	0
10	daikibo-berlin	LaserCutter	0

Key Insight

Breakdowns are factory-specific not equipment-wide. LaserWelder fails only in Seiko (48), LaserCutter fails only in Shenzhen (39) and Meiyo(4). The same machine types perform perfectly in other locations. This rules out global equipment defects and points to localised maintenance or operational failures.

Query 9: Temperature Correlation — Avg Temperature During Healthy vs Unhealthy Status

Business Question: *Is there a measurable correlation between machine operating temperature and breakdown events — do machines run significantly hotter when they fail?*

	factory text	status text	avg_temperature numeric
1	daikibo-berlin	healthy	24.99
2	daikibo-berlin	unhealthy	25.00
3	daikibo-factory-meiyo	healthy	24.98
4	daikibo-factory-meiyo	unhealthy	25.18
5	daikibo-factory-seiko	healthy	25.00
6	daikibo-factory-seiko	unhealthy	24.77
7	daikibo-shenzhen	healthy	24.99
8	daikibo-shenzhen	unhealthy	24.93

Key Insight

No meaningful temperature correlation found. Average temperature difference between healthy and unhealthy status is less than 0.5 degrees Celsius across all factories. Breakdowns are not heat-induced — they are mechanical or process-related. This null finding narrows the root cause investigation significantly.

Query 10: Daily Breakdown Trend — Peak Failure Days in May 2021

Business Question: *Which specific days in May 2021 recorded the highest breakdown counts — and is there a temporal pattern that could inform future maintenance scheduling?*

	day_in_month numeric	breakdowns bigint
1	5	48
2	2	41
3	3	9
4	7	2
5	1	1
6	8	1
7	4	1
8	9	0
9	6	0
10	10	0

Key Insight

89% of all breakdowns occurred in the first 5 days of May 2021 — Day 5 alone recorded 48 and Day 2 recorded 41. After Day 7 breakdowns dropped to near zero. This infant mortality pattern suggests machines failed shortly after a restart or new deployment — and a maintenance intervention likely resolved the issue by mid-month.